JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Sequences and Series of Real Numbers (2005–2026 Archive)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
131	292 M	393 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	85	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	15	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	31	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q43 • ID: JAM_2026_Q43) NAT +1 M Neg: Nil (0)

Q.43	$\lim_{n\to\infty}\left(\frac{n^2+1}{\sqrt{n^6+1}}+\cdots+\frac{n^2+n}{\sqrt{n^6+n}}\right)=$_____(*rounded off to one decimal place*).

Question 2 (JAM 2026 • Q42 • ID: JAM_2026_Q42) NAT +1 M Neg: Nil (0)

Q.42	$\int_0^1\left(\sum_{k=1}^{\infty}\frac{(\log_e 2)^k x^{k-1}}{(k-1)!}\right)dx =$ _____(*rounded off to one decimal place*).

Question 3 (JAM 2026 • Q41 • ID: JAM_2026_Q41) NAT +1 M Neg: Nil (0)

Q.41	The radius of convergence of the series $\sum_{n=1}^{\infty}\frac{(n!)^2}{(2n)!}(\log_e n)^{-1}x^n$ is _____ (*rounded off to one decimal place*).

Question 4 (JAM 2026 • Q32 • ID: JAM_2026_Q32) MSQ +2 M Neg: Nil (0)

Q.32	Which of the following statements is/are TRUE?
(A)	There exists a monotone sequence that does not converge but has a convergent subsequence.
(B)	There exists a sequence that has a bounded subsequence but does not have any convergent subsequence.
(C)	There exists a sequence (x_n) such that given any positive integer m, (x_n) has a subsequence converging to m.
(D)	There exists a sequence (x_n) such that $(\lvert x_{n+1} - x_n \rvert)$ converges to 0 but (x_n) does not converge.

Question 5 (JAM 2026 • Q29 • ID: JAM_2026_Q29) MCQ +2 M Neg: −0.67

Q.29	For each positive integer n, let $$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log_e n$$ and $$y_n = \int_1^n \frac{\cos t}{t^2}\, dt\ .$$ Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is convergent and (y_n) is not convergent.
(C)	(x_n) is not convergent and (y_n) is convergent.
(D)	(x_n) is not convergent and (y_n) is not convergent.

Question 6 (JAM 2026 • Q12 • ID: JAM_2026_Q12) MCQ +2 M Neg: −0.67

Q.12	For which one of the following pairs of values of a and b, does the series $\sum_{k=1}^{\infty} \frac{(3x)^k}{2^{\sqrt{k}}(1-x)^k}$ converge for all $x \in (a, b)$?
(A)	$a = \frac{-1}{4}$ and $b = \frac{1}{2}$
(B)	$a = -1$ and $b = \frac{1}{5}$
(C)	$a = \frac{-1}{2}$ and $b = \frac{1}{4}$
(D)	$a = \frac{-1}{4}$ and $b = \frac{1}{3}$

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Question 7 (JAM 2026 • Q11 • ID: JAM_2026_Q11) MCQ +2 M Neg: −0.67

Q.11	Given a sequence of real numbers (a_n), define $b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0 \end{cases}$ and $c_n = \begin{cases} a_n, & \text{if } a_n < 0 \\ 0, & \text{if } a_n \geq 0 \end{cases}$ for each positive integer n. Which ONE of the following statements is TRUE?
(A)	If $\sum_{n=1}^{\infty} a_n$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(B)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} \lvert a_n \rvert$ does not converge, then $\sum_{n=1}^{\infty} b_n$ converges.
(C)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} \lvert a_n \rvert$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(D)	If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges or $\sum_{n=1}^{\infty} c_n$ converges.

Question 8 (JAM 2026 • Q2 • ID: JAM_2026_Q2) MCQ +1 M Neg: −0.33

Q.2	For each positive integer n, define $x_n = (-1)^n$. Which ONE of the following statements about the sequence (x_n) is FALSE?
(A)	There exists $\epsilon > 0$ such that $\lvert x_n - 1\rvert < \epsilon$ for all positive integers n.
(B)	There exists $\epsilon > 0$ such that for all $M > 0$ there exists a positive integer $N > M$ for which $\lvert x_N - 1\rvert > \epsilon$.
(C)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $\lvert x_N - 1\rvert < \epsilon$.
(D)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $\lvert x_N - 1\rvert > \epsilon$.

Question 9 (JAM 2026 • Q1 • ID: JAM_2026_Q1) MCQ +1 M Neg: −0.33

Q.1	For each positive integer n, let $x_n = 1 - (-1)^n + \frac{1}{n}$ and $y_n = \left(1 + \frac{1}{2n}\right)^{3n}$. Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is not convergent and (y_n) is not convergent.
(C)	(x_n) is convergent and (y_n) is not convergent.
(D)	(x_n) is not convergent and (y_n) is convergent.

Question 10 (JAM 2025 • Q54 • ID: JAM_2025_Q54) NAT +2 M Neg: Nil (0)

Q.54 The value of the infinite series

$$\sum_{n=1}^{\infty} n \left(\frac{3}{4}\right)^{2(n-1)}$$

is equal to _____ *(rounded off to two decimal places)*

Question 11 (JAM 2025 • Q41 • ID: JAM_2025_Q41) NAT +1 M Neg: Nil (0)

Q.41 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{\left(x+\frac{1}{4}\right)^n}{(-2)^n n^2}$$

about $x = -\frac{1}{4}$, is equal to _____ *(rounded off to two decimal places).*

Question 12 (JAM 2025 • Q36 • ID: JAM_2025_Q36) MSQ +2 M Neg: Nil (0)

Q.36 For $n \in \mathbb{N}$, let

$$x_n = \sum_{k=1}^{n} \frac{k}{n^2 + k}.$$

Then, which of the following is/are TRUE?

(A) The sequence (x_n) converges

(B) The series $\sum_{n=1}^{\infty} x_n$ converges

(C) The series $\sum_{n=1}^{\infty} x_n$ does NOT converge

(D) The series $\sum_{n=1}^{\infty} x_n^n$ converges

Question 13 (JAM 2025 • Q26 • ID: JAM_2025_Q26) MCQ +2 M Neg: −0.67

Q.26 Let $x_1 > 0$. For $n \in \mathbb{N}$, define $x_{n+1} = x_n + 4$. If

$$\lim_{n \to \infty} \left(\frac{1}{x_2 x_3} + \frac{1}{x_3 x_4} + \cdots + \frac{1}{x_{n+1} x_{n+2}} \right) = \frac{1}{24},$$

then the value of x_1 is equal to

(A) 1

(B) 2

(C) 3

(D) 8

Question 14 (JAM 2025 • Q25 • ID: JAM_2025_Q25) MCQ +2 M Neg: −0.67

Q.25 Let $x_1 = 1$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \left(\frac{1}{2} + \frac{\sin^2 n}{n}\right) x_n .$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} x_n$ converges

(B) $\sum_{n=1}^{\infty} x_n$ does NOT converge

(C) $\sum_{n=1}^{\infty} x_n^2$ does NOT converge

(D) $\sum_{n=1}^{\infty} x_n x_{n+1}$ does NOT converge

Question 15 (JAM 2025 • Q21 • ID: JAM_2025_Q21) MCQ +2 M Neg: −0.67

Q.21 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \frac{3^n}{n^3} \text{ and } y_n = (4^n + (-1)^n 3^n)^{1/n} .$$

Then, which one of the following is TRUE?

(A) (x_n) has a convergent subsequence, and NO subsequence of (y_n) is convergent

(B) NO subsequence of (x_n) is convergent, and (y_n) has a convergent subsequence

(C) (x_n) has a convergent subsequence, and (y_n) has a convergent subsequence

(D) NO subsequence of (x_n) is convergent, and NO subsequence of (y_n) is convergent

Question 16 (JAM 2025 • Q20 • ID: JAM_2025_Q20) MCQ +2 M Neg: −0.67

Q.20 Let $x_1 = 2$ and $x_{n+1} = 2 + \frac{1}{2x_n}$ for all $n \in \mathbb{N}$.

Then, which one of the following is TRUE?

(A) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence

(B) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence

(C) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

(D) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

Question 17 (JAM 2025 • Q19 • ID: JAM_2025_Q19) MCQ +2 M Neg: −0.67

Q.19 Let $x_1 = \frac{5}{2}$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \frac{1}{5}(x_n^2 + 6).$$

Then, which one of the following is TRUE?

(A) (x_n) is an increasing sequence, and (x_n) is NOT a bounded sequence

(B) (x_n) is NOT an increasing sequence, and (x_n) is NOT a bounded sequence

(C) (x_n) is NOT a decreasing sequence, and (x_n) is a bounded sequence

(D) (x_n) is a decreasing sequence, and (x_n) is a bounded sequence

Question 18 (JAM 2025 • Q18 • ID: JAM_2025_Q18) MCQ +2 M Neg: −0.67

Q.18 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \cos\frac{1}{n} \quad \text{and} \quad y_n = \sum_{k=1}^{n} \frac{1}{n+k}.$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

(B) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ converges

(C) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ converges

(D) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

Question 19 (JAM 2025 • Q1 • ID: JAM_2025_Q1) MCQ +1 M Neg: −0.33

Q.1 The sum of the infinite series

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\pi^{2n+1}}{2^{2n+1}(2n)!}$$

is equal to

(A) $-\pi$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $-\frac{\pi}{4}$

Question 20 (JAM 2024 • Q59 • ID: JAM_2024_Q59) NAT +2 M Neg: Nil (0)

Q.59 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{n^{n-1}}\sum_{k=0}^{n}\frac{n!}{k!\,(n-k)!}\,\frac{n^k}{k+1}$$

and $\beta = \lim_{n\to\infty} a_n$. Then, the value of $\log\beta$ equals _____ (*rounded off to two decimal places*).

Question 21 (JAM 2024 • Q55 • ID: JAM_2024_Q55) NAT +2 M Neg: Nil (0)

Q.55 For $n \in \mathbb{N}$, if

$$a_n = \frac{1}{n^3+1} + \frac{2^2}{n^3+2} + \cdots + \frac{n^2}{n^3+n}$$

then the sequence $\{a_n\}_{n=1}^{\infty}$ converges to _____ (*rounded off to two decimal places*)

Question 22 (JAM 2024 • Q47 • ID: JAM_2024_Q47) NAT +1 M Neg: Nil (0)

Q.47 Let $a_1 = 1, b_1 = 2$ and $c_1 = 3$. Consider the convergent sequences

$$\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty} \text{ and } \{c_n\}_{n=1}^{\infty}$$

defined as follows:

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{b_n + c_n}{2} \text{ and } c_{n+1} = \frac{c_n + a_n}{2} \text{ for } n \geq 1.$$

Then,

$$\sum_{n=1}^{\infty} b_n c_n (a_{n+1} - a_n) + \sum_{n=1}^{\infty} (b_{n+1}c_{n+1} - b_n c_n) a_{n+1}$$

equals _____ (*rounded off to two decimal places*)

Question 23 (JAM 2024 • Q39 • ID: JAM_2024_Q39) MSQ +2 M Neg: Nil (0)

Q.39 For $0 < \alpha < 4$, define the sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers as follows:

$$x_1 = \alpha \text{ and } x_{n+1} + 2 = -x_n(x_n - 4) \text{ for } n \in \mathbb{N}.$$

Which of the following is/are TRUE?

(A) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (0,1)$

(B) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (1,2)$

(C) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (2,3)$

(D) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (3,4)$

Question 24 (JAM 2024 • Q32 • ID: JAM_2024_Q32) MSQ +2 M Neg: Nil (0)

Q.32 Which of the following statements is/are TRUE?

(A) $\sum_{n=1}^{\infty} n \log\left(1 + \frac{1}{n^3}\right)$ is convergent

(B) $\sum_{n=1}^{\infty} \left(1 - \cos\left(\frac{1}{n}\right)\right) \log n$ is convergent

(C) $\sum_{n=1}^{\infty} n^2 \log\left(1 + \frac{1}{n^3}\right)$ is convergent

(D) $\sum_{n=1}^{\infty} \left(1 - \cos\left(\frac{1}{\sqrt{n}}\right)\right) \log n$ is convergent

Question 25 (JAM 2024 • Q31 • ID: JAM_2024_Q31) MSQ +2 M Neg: Nil (0)

Q.31 Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers.

Then, which of the following statements is/are always TRUE?

(A) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^2$ converges absolutely

(B) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^3$ converges absolutely

(C) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges

(D) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^3$ converges

Madras

Question 26 (JAM 2024 • Q28 • ID: JAM_2024_Q28) MCQ +2 M Neg: −0.67

Q.28 For $n \geq 3$, let a regular n-sided polygon P_n be circumscribed by a circle of radius R_n and let r_n be the radius of the circle inscribed in P_n. Then

$$\lim_{n\to\infty}\left(\frac{R_n}{r_n}\right)^{n^2}$$

equals

(A) $e^{(\pi^2)}$

(B) $e^{\left(\frac{\pi^2}{2}\right)}$

(C) $e^{\left(\frac{\pi^2}{3}\right)}$

(D) $e^{(2\pi^2)}$

Madras

Question 27 (JAM 2024 • Q21 • ID: JAM_2024_Q21) MCQ +2 M Neg: −0.67

Q.21 For $p, q, r \in \mathbb{R}$, $r \neq 0$ and $n \in \mathbb{N}$, let

$$a_n = p^n\, n^q \left(\frac{n}{n+2}\right)^{n^2} \text{ and } b_n = \frac{n^n}{n!\; r^n}\left(\sqrt{\frac{n+2}{n}}\right).$$

Then, which one of the following statements is TRUE?

(A) If $1 < p < e^2$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent

(B) If $e^2 < p < e^4$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent

(C) If $1 < r < e$, then $\sum_{n=1}^{\infty} b_n$ is convergent

(D) If $\frac{1}{e} < r < 1$, then $\sum_{n=1}^{\infty} b_n$ is convergent

Madras

Question 28 (JAM 2024 • Q20 • ID: JAM_2024_Q20) MCQ +2 M Neg: −0.67

Q.20 Define the sequences $\{a_n\}_{n=3}^{\infty}$ and $\{b_n\}_{n=3}^{\infty}$ as

$$a_n = (\log n + \log\log n)^{\log n} \quad \text{and} \quad b_n = n^{\left(1+\frac{1}{\log n}\right)}.$$

Which one of the following is TRUE?

(A) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is convergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is divergent

(B) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is divergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is convergent

(C) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are divergent

(D) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are convergent

Madras

Question 29 (JAM 2024 • Q14 • ID: JAM_2024_Q14) MCQ +2 M Neg: −0.67

Q.14 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{(3n+2)(3n+4)} \quad \text{and} \quad b_n = \frac{n^3 + \cos(3^n)}{3^n + n^3}.$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is divergent

(B) $\sum_{n=1}^{\infty} a_n$ is divergent but $\sum_{n=1}^{\infty} b_n$ is convergent

(C) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent

(D) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

Madras

Question 30 (JAM 2023 • Q50 • ID: JAM_2023_Q50) NAT +1 M Neg: Nil (0)

Q. 50 The value of $\lim_{n\to\infty} \left(1 + \frac{1}{2^n} + \frac{1}{3^n} + \cdots + \frac{1}{(2023)^n}\right)^{\frac{1}{n}}$ is equal to _____.
(rounded off to two decimal places)

Question 31 (JAM 2023 • Q49 • ID: JAM_2023_Q49) NAT +1 M Neg: Nil (0)

Q. 49 The sum of the series $\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}$ is equal to _____.
(rounded off to two decimal places)

Question 32 (JAM 2023 • Q40 • ID: JAM_2023_Q40) MSQ +2 M Neg: Nil (0)

Q. 40 Let R_1 and R_2 be the radii of convergence of the power series $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ and $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$, respectively. Then

(A) $R_1 = R_2$

(B) $R_2 > 1$

(C) $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ converges for all $x \in [-1, 1]$

(D) $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$ converges for all $x \in [-1, 1]$

Question 33 (JAM 2023 • Q24 • ID: JAM_2023_Q24) MCQ +2 M Neg: −0.67

Q. 24 Suppose $f : (-1, 1) \to \mathbb{R}$ is an infinitely differentiable function such that the series $\sum_{j=0}^{\infty} a_j \frac{x^j}{j!}$ converges to $f(x)$ for each $x \in (-1, 1)$, where,

$$a_j = \int_{0}^{\pi/2} \theta^j \cos^j(\tan\theta) d\theta + \int_{\pi/2}^{\pi} (\theta - \pi)^j \cos^j(\tan\theta) d\theta$$

for $j \geq 0$. Then

(A) $f(x) = 0$ for all $x \in (-1, 1)$

(B) f is a non-constant even function on $(-1, 1)$

(C) f is a non-constant odd function on $(-1, 1)$

(D) f is NEITHER an odd function NOR an even function on $(-1, 1)$

Question 34 (JAM 2023 • Q21 • ID: JAM_2023_Q21) MCQ +2 M Neg: −0.67

Q. 21 Let (a_n) and (b_n) be sequences of real numbers such that

$$|a_n - a_{n+1}| = \frac{1}{2^n} \quad \text{and} \quad |b_n - b_{n+1}| = \frac{1}{\sqrt{n}} \quad \text{for } n \in \mathbb{N}.$$

Then

(A) both (a_n) and (b_n) are Cauchy sequences

(B) (a_n) is a Cauchy sequence but (b_n) need NOT be a Cauchy sequence

(C) (a_n) need NOT be a Cauchy sequence but (b_n) is a Cauchy sequence

(D) both (a_n) and (b_n) need NOT be Cauchy sequences

Question 35 (JAM 2023 • Q20 • ID: JAM_2023_Q20) MCQ +2 M Neg: −0.67

Q. 20 Let (a_n) be a sequence of real numbers such that the series $\sum_{n=0}^{\infty} a_n(x-2)^n$ converges at $x = -5$. Then this series also converges at

(A) $x = 9$

(B) $x = 12$

(C) $x = 5$

(D) $x = -6$

Question 36 (JAM 2023 • Q19 • ID: JAM_2023_Q19) MCQ +2 M Neg: −0.67

Q. 19 Let $a_n = \dfrac{1 + 2^{-2} + \cdots + n^{-2}}{n}$ for $n \in \mathbb{N}$. Then

(A) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are convergent

(B) the sequence (a_n) is convergent but the series $\sum_{n=1}^{\infty} a_n$ is NOT convergent

(C) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are NOT convergent

(D) the sequence (a_n) is NOT convergent but the series $\sum_{n=1}^{\infty} a_n$ is convergent

Question 37 (JAM 2023 • Q11 • ID: JAM_2023_Q11) MCQ +2 M Neg: −0.67

Q. 11 Let $a_n = \sin\left(\dfrac{1}{n^3}\right)$ and $b_n = \sin\left(\dfrac{1}{n}\right)$ for $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

(B) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is NOT convergent

(C) $\sum_{n=1}^{\infty} a_n$ is NOT convergent but $\sum_{n=1}^{\infty} b_n$ is convergent

(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are NOT convergent

Question 38 (JAM 2023 • Q10 • ID: JAM_2023_Q10) MCQ +1 M Neg: −0.33

Q. 10 Let (a_n) be a sequence of real numbers defined by

$$a_n = \begin{cases} 1 & \text{if } n \text{ is prime} \\ -1 & \text{if } n \text{ is not prime.} \end{cases}$$

Let $b_n = \dfrac{a_n}{n}$ for $n \in \mathbb{N}$. Then

(A) both (a_n) and (b_n) are convergent

(B) (a_n) is convergent but (b_n) is NOT convergent

(C) (a_n) is NOT convergent but (b_n) is convergent

(D) both (a_n) and (b_n) are NOT convergent

Question 39 (JAM 2023 • Q9 • ID: JAM_2023_Q9) MCQ +1 M Neg: −0.33

Q. 9 Let $a_n = \left(1 + \dfrac{1}{n}\right)^n$ and $b_n = n \cos\left(\dfrac{n!\pi}{2^{10}}\right)$ for $n \in \mathbb{N}$. Then

(A) (a_n) is convergent and (b_n) is bounded

(B) (a_n) is NOT convergent and (b_n) is bounded

(C) (a_n) is convergent and (b_n) is unbounded

(D) (a_n) is NOT convergent and (b_n) is unbounded

Question 40 (JAM 2022 • Q55 • ID: JAM_2022_Q55) NAT +2 M Neg: Nil (0)

Q.55	Let r be the radius of convergence of the power series $$\frac{1}{3}+\frac{x}{5}+\frac{x^2}{3^2}+\frac{x^3}{5^2}+\frac{x^4}{3^3}+\frac{x^5}{5^3}+\frac{x^6}{3^4}+\frac{x^7}{5^4}+\cdots.$$ Then the value of r^2 is equal to _____. (*Rounded off to two decimal places*)

Question 41 (JAM 2022 • Q44 • ID: JAM_2022_Q44) NAT +1 M Neg: Nil (0)

Q.44	The sum of the series $$\sum_{n=1}^{\infty}\frac{1}{(4n-3)(4n+1)}$$ is equal to _____. (*Rounded off to two decimal places*)

Question 42 (JAM 2022 • Q38 • ID: JAM_2022_Q38) MSQ +2 M Neg: Nil (0)

Q.38	Let (x_n) be a sequence of real numbers. Consider the set $$P = \{n \in \mathbb{N} : x_n > x_m \text{ for all } m \in \mathbb{N} \text{ with } m > n\}.$$ Then which of the following is/are true?
(A)	If P is finite, then (x_n) has a monotonically increasing subsequence.
(B)	If P is finite, then no subsequence of (x_n) is monotonically increasing.
(C)	If P is infinite, then (x_n) has a monotonically decreasing subsequence.
(D)	If P is infinite, then no subsequence of (x_n) is monotonically decreasing.

Question 43 (JAM 2022 • Q28 • ID: JAM_2022_Q28) MCQ +2 M Neg: −0.67

Q.28	For $-1 \le x \le 1$, if $f(x)$ is the sum of the convergent power series $$x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \cdots + \frac{x^n}{n^2} + \cdots$$ then $f\left(\frac{1}{2}\right)$ is equal to
(A)	$\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\, dt.$
(B)	$-\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\, dt.$
(C)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1+t)\, dt.$
(D)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1-t)\, dt.$

Question 44 (JAM 2022 • Q14 • ID: JAM_2022_Q14) MCQ +2 M Neg: −0.67

Q.14	The value of $$\lim_{n\to\infty}\sum_{k=2}^{n}\frac{\sqrt{n+1}-\sqrt{n}}{k(\ln k)^2}$$ is equal to
(A)	∞
(B)	1
(C)	e
(D)	0

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Question 45 (JAM 2022 • Q12 • ID: JAM_2022_Q12) MCQ +2 M Neg: −0.67

Q.12	Let T denote the sum of the convergent series $$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\cdots+\frac{(-1)^{n+1}}{n}+\cdots$$ and let S denote the sum of the convergent series $$1-\frac{1}{2}-\frac{1}{4}+\frac{1}{3}-\frac{1}{6}-\frac{1}{8}+\frac{1}{5}-\frac{1}{10}-\frac{1}{12}+\cdots=\sum_{n=1}^{\infty}a_n,$$ where $$a_{3m-2}=\frac{1}{2m-1},\quad a_{3m-1}=\frac{-1}{4m-2}\quad\text{and}\quad a_{3m}=\frac{-1}{4m}\quad\text{for } m\in\mathbb{N}.$$ Then which one of the following is true?
(A)	$T=S$ and $S\neq 0$.
(B)	$2T=S$ and $S\neq 0$.
(C)	$T=2S$ and $S\neq 0$.
(D)	$T=S=0$.

Question 46 (JAM 2022 • Q8 • ID: JAM_2022_Q8) MCQ +1 M Neg: −0.33

Q.8	Consider the series $$\sum_{n=1}^{\infty} \frac{1}{n^m\left(1+\frac{1}{n^p}\right)}$$ where m and p are real numbers. Under which of the following conditions does the above series converge?
(A)	$m > 1$.
(B)	$0 < m < 1$ and $p > 1$.
(C)	$0 < m \leq 1$ and $0 \leq p \leq 1$.
(D)	$m = 1$ and $p > 1$.

Question 47 (JAM 2022 • Q7 • ID: JAM_2022_Q7) MCQ +1 M Neg: −0.33

Q.7	Suppose $$a_n = \frac{3^n + 3}{5^n - 5} \quad \text{and} \quad b_n = \frac{1}{(1+n^2)^{\frac{1}{4}}} \quad \text{for } n = 2,3,4,\ldots.$$ Then which one of the following is true?
(A)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are convergent.
(B)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are divergent.
(C)	$\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is divergent.
(D)	$\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is convergent.

Question 48 (JAM 2022 • Q6 • ID: JAM_2022_Q6) MCQ +1 M Neg: −0.33

Q.6	Let (x_n) and (y_n) be sequences of real numbers defined by $x_1 = 1,\ \ y_1 = \frac{1}{2},\ \ x_{n+1} = \frac{x_n + y_n}{2},\ \ \text{and}\ \ y_{n+1} = \sqrt{x_n y_n}\ \ \text{for all}\ n \in \mathbb{N}.$ Then which one of the following is true?
(A)	(x_n) is convergent, but (y_n) is not convergent.
(B)	(x_n) is not convergent, but (y_n) is convergent.
(C)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n > \lim_{n\to\infty} y_n$.
(D)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n$.

Question 49 (JAM 2022 • Q5 • ID: JAM_2022_Q5) MCQ +1 M Neg: −0.33

Q.5	The radius of convergence of the power series $$\sum_{n=1}^{\infty}\left(\frac{n^3}{4^n}\right)x^{5n}$$ is
(A)	4
(B)	$\sqrt[5]{4}$
(C)	$\frac{1}{4}$
(D)	$\frac{1}{\sqrt[5]{4}}$

Question 50 (JAM 2021 • Q55 • ID: JAM_2021_Q55) NAT +2 M Neg: Nil (0)

Q. 55 Define the sequence

$$s_n = \begin{cases} \frac{1}{2^n}\sum_{j=0}^{n-2} 2^{2j} & \text{if } n > 0 \text{ is even,} \\ \frac{1}{2^n}\sum_{j=0}^{n-1} 2^{2j} & \text{if } n > 0 \text{ is odd.} \end{cases}$$

Define $\sigma_m = \frac{1}{m}\sum_{n=1}^{m} s_n$. The number of limit points of the sequence $\{\sigma_m\}$ is _____.

Question 51 (JAM 2021 • Q50 • ID: JAM_2021_Q50) NAT +1 M Neg: Nil (0)

Q. 50 The value of

$$\frac{\pi}{2}\lim_{n\to\infty}\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cdots\cos\left(\frac{\pi}{2^{n+1}}\right)$$

is _____.

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Question 52 (JAM 2021 • Q42 • ID: JAM_2021_Q42) NAT +1 M Neg: Nil (0)

Q. 42 The value of

$$\lim_{n\to\infty}\left(3^n+5^n+7^n\right)^{\frac{1}{n}}$$

is _____.

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Question 53 (JAM 2021 • Q30 • ID: JAM_2021_Q30) MCQ +2 M Neg: −0.67

Q. 30 Consider the two series

$$\text{I.} \quad \sum_{n=1}^{\infty} \frac{1}{n^{1+(1/n)}} \quad \text{and} \quad \text{II.} \quad \sum_{n=1}^{\infty} \frac{1}{n^{2-n^{1/n}}}$$

Which one of the following holds?

(A) Both I and II converge.

(B) Both I and II diverge.

(C) I converges and II diverges.

(D) I diverges and II converges.

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Question 54 (JAM 2021 • Q15 • ID: JAM_2021_Q15) MCQ +2 M Neg: −0.67

JAM 2021 MATHEMATICS - MA

Q. 15 Let $f : \mathbb{N} \to \mathbb{N}$ be a bijective map such that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty.$$

The number of such bijective maps is

(A) exactly one.

(B) zero.

(C) finite but more than one.

(D) infinite.

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Question 55 (JAM 2020 • Q52 • ID: JAM_2020_Q52) NAT +2 M Neg: Nil (0)

Q. 52 Consider the expansion of the function $f(x) = \dfrac{3}{(1-x)(1+2x)}$ in powers of x, that is valid in $|x| < \dfrac{1}{2}$. Then the coefficient of x^4 is _____.

Question 56 (JAM 2020 • Q51 • ID: JAM_2020_Q51) NAT +2 M Neg: Nil (0)

Q. 51 The sum of the series $\dfrac{1}{2(2^2-1)} + \dfrac{1}{3(3^2-1)} + \dfrac{1}{4(4^2-1)} + \cdots$ is _____.

Question 57 (JAM 2020 • Q41 • ID: JAM_2020_Q41) NAT +1 M Neg: Nil (0)

Q. 41 Let $x_n = n^{\frac{1}{n}}$ and $y_n = e^{1-x_n}, n \in \mathbb{N}$. Then the value of $\lim\limits_{n\to\infty} y_n$ is _____.

Question 58 (JAM 2020 • Q39 • ID: JAM_2020_Q39) MSQ +2 M Neg: Nil (0)

Q. 39 Let f be a real valued function of a real variable, such that $|f^{(n)}(0)| \leq K$ for all $n \in \mathbb{N}$, where $K > 0$. Which of the following is/are true?

(A) $\left|\frac{f^{(n)}(0)}{n!}\right|^{\frac{1}{n}} \to 0$ as $n \to \infty$

(B) $\left|\frac{f^{(n)}(0)}{n!}\right|^{\frac{1}{n}} \to \infty$ as $n \to \infty$

(C) $f^{(n)}(x)$ exists for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$

(D) The series $\sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{(n-1)!}$ is absolutely convergent

Question 59 (JAM 2020 • Q37 • ID: JAM_2020_Q37) MSQ +2 M Neg: Nil (0)

JAM 2020 MATHEMATICS - MA

Q. 37 Let $a = \lim_{n\to\infty}\left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{(n-1)}{n^2}\right)$ and $b = \lim_{n\to\infty}\left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n}\right)$. Which of the following is/are true?

(A) $a > b$ (B) $a < b$ (C) $ab = \ln\sqrt{2}$ (D) $\frac{a}{b} = \ln\sqrt{2}$

Question 60 (JAM 2020 • Q32 • ID: JAM_2020_Q32) MSQ +2 M Neg: Nil (0)

Q. 32 If $s_n = \frac{(-1)^n}{2^n+3}$ and $t_n = \frac{(-1)^n}{4n-1}, n = 0, 1, 2, ...$, then

(A) $\sum_{n=0}^{\infty} s_n$ is absolutely convergent

(B) $\sum_{n=0}^{\infty} t_n$ is absolutely convergent

(C) $\sum_{n=0}^{\infty} s_n$ is conditionally convergent

(D) $\sum_{n=0}^{\infty} t_n$ is conditionally convergent

Question 61 (JAM 2020 • Q13 • ID: JAM_2020_Q13) MCQ +2 M Neg: −0.67

JAM 2020 MATHEMATICS - MA

Q. 13 Suppose that S is the sum of a convergent series $\sum_{n=1}^{\infty} a_n$. Define $t_n = a_n + a_{n+1} + a_{n+2}$. Then the series $\sum_{n=1}^{\infty} t_n$

(A) diverges

(B) converges to $3S - a_1 - a_2$

(C) converges to $3S - a_1 - 2a_2$

(D) converges to $3S - 2a_1 - a_2$

Question 62 (JAM 2020 • Q12 • ID: JAM_2020_Q12) MCQ +2 M Neg: −0.67

Q. 12 Define $s_1 = \alpha > 0$ and $s_{n+1} = \sqrt{\dfrac{1+s_n^2}{1+\alpha}}$, $n \geq 1$. Which of the following is true?

(A) If $s_n^2 < \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\sqrt{\alpha}}$

(B) If $s_n^2 < \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\alpha}$

(C) If $s_n^2 > \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\sqrt{\alpha}}$

(D) If $s_n^2 > \dfrac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n\to\infty} s_n = \dfrac{1}{\alpha}$

Question 63 (JAM 2020 • Q11 • ID: JAM_2020_Q11) MCQ +2 M Neg: −0.67

Q. 11 Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $l = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$. Which of the following is true?

(A) If $l = 1$, then $\lim_{n\to\infty} a_n = 1$

(B) If $l = 1$, then $\lim_{n\to\infty} a_n = 0$

(C) If $l < 1$, then $\lim_{n\to\infty} a_n = 1$

(D) If $l < 1$, then $\lim_{n\to\infty} a_n = 0$

Question 64 (JAM 2020 • Q9 • ID: JAM_2020_Q9) MCQ +1 M Neg: −0.33

Q. 9 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \left(\frac{n+2}{n}\right)^{n^2} x^n$$

is

(A) e^2 (B) $\frac{1}{\sqrt{e}}$ (C) $\frac{1}{e}$ (D) $\frac{1}{e^2}$

Question 65 (JAM 2020 • Q1 • ID: JAM_2020_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let $s_n = 1 + \frac{(-1)^n}{n}, n \in \mathbb{N}$. Then the sequence $\{s_n\}$ is

(A) monotonically increasing and is convergent to 1

(B) monotonically decreasing and is convergent to 1

(C) neither monotonically increasing nor monotonically decreasing but is convergent to 1

(D) divergent

Question 66 (JAM 2019 • Q52 • ID: JAM_2019_Q52) NAT +2 M Neg: Nil (0)

Q.52 The greatest lower bound of the set

$$\{(e^n + 2^n)^{\frac{1}{n}} : n \in \mathbb{N}\},$$

(round off to 2 decimal places) is _____

Question 67 (JAM 2019 • Q40 • ID: JAM_2019_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let $\{a_n\}$ be the sequence of real numbers such that

$$a_1 = 1 \text{ and } a_{n+1} = a_n + a_n^2 \text{ for all } n \geq 1.$$

Then

(A) $a_4 = a_1(1 + a_1)(1 + a_2)(1 + a_3)$

(B) $\lim_{n\to\infty} \frac{1}{a_n} = 0$

(C) $\lim_{n\to\infty} \frac{1}{a_n} = 1$

(D) $\lim_{n\to\infty} a_n = 0$

Question 68 (JAM 2019 • Q36 • ID: JAM_2019_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Let $\{a_n\}$ be the sequence given by

$$a_n = \max\left\{\sin\left(\frac{n\pi}{3}\right), \cos\left(\frac{n\pi}{3}\right)\right\}, \quad n \geq 1.$$

Then which of the following statements is/are TRUE about the subsequences $\{a_{6n-1}\}$ and $\{a_{6n+4}\}$?

(A) Both the subsequences are convergent

(B) Only one of the subsequences is convergent

(C) $\{a_{6n-1}\}$ converges to $-\frac{1}{2}$

(D) $\{a_{6n+4}\}$ converges to $\frac{1}{2}$

Question 69 (JAM 2019 • Q30 • ID: JAM_2019_Q30) MCQ +2 M Neg: −0.67

Q.30 Let $\{a_n\}$ be a sequence of positive real numbers such that

$$a_1 = 1,\ a_{n+1}^2 - 2a_n a_{n+1} - a_n = 0 \text{ for all } n \geq 1.$$

Then the sum of the series $\sum_{n=1}^{\infty} \frac{a_n}{3^n}$ lies in the interval

(A) $(1, 2]$ (B) $(2, 3]$ (C) $(3, 4]$ (D) $(4, 5]$

Question 70 (JAM 2019 • Q28 • ID: JAM_2019_Q28) MCQ +2 M Neg: −0.67

Q.28 Let $\{a_n\}$ be a sequence of positive real numbers. The series $\sum_{n=1}^{\infty} a_n$ converges if the series

(A) $\sum_{n=1}^{\infty} a_n^2$ converges
(B) $\sum_{n=1}^{\infty} \frac{a_n}{2^n}$ converges
(C) $\sum_{n=1}^{\infty} \frac{a_{n+1}}{a_n}$ converges
(D) $\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}}$ converges

Question 71 (JAM 2019 • Q19 • ID: JAM_2019_Q19) MCQ +2 M Neg: −0.67

Q.19 Which one of the following series is divergent?

(A) $\sum_{n=1}^{\infty} \frac{1}{n} \sin^2 \frac{1}{n}$

(B) $\sum_{n=1}^{\infty} \frac{1}{n} \log n$

(C) $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{1}{n}$

(D) $\sum_{n=1}^{\infty} \frac{1}{n} \tan \frac{1}{n}$

Question 72 (JAM 2019 • Q15 • ID: JAM_2019_Q15) MCQ +2 M Neg: −0.67

Q.15 For $-1 < x < 1$, the sum of the power series $1 + \sum_{n=2}^{\infty} (-1)^{n-1} n^2 x^{n-1}$ is

(A) $\frac{1-x}{(1+x)^3}$

(B) $\frac{1+x^2}{(1+x)^4}$

(C) $\frac{1-x}{(1+x)^2}$

(D) $\frac{1+x^2}{(1+x)^3}$

Question 73 (JAM 2019 • Q4 • ID: JAM_2019_Q4) MCQ +1 M Neg: −0.33

Q.4 Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be sequences of positive real numbers such that $na_n < b_n < n^2 a_n$ for all $n \geq 2$. If the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$ is 4, then the power series $\sum_{n=0}^{\infty} b_n x^n$

(A) converges for all x with $|x| < 2$
(B) converges for all x with $|x| > 2$
(C) does not converge for any x with $|x| > 2$
(D) does not converge for any x with $|x| < 2$

Question 74 (JAM 2019 • Q1 • ID: JAM_2019_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Let $a_1 = b_1 = 0$, and for each $n \geq 2$, let a_n and b_n be real numbers given by

$$a_n = \sum_{m=2}^{n} \frac{(-1)^m m}{(\log(m))^m} \text{ and } b_n = \sum_{m=2}^{n} \frac{1}{(\log(m))^m}.$$

Then which one of the following is TRUE about the sequences $\{a_n\}$ and $\{b_n\}$?

(A) Both $\{a_n\}$ and $\{b_n\}$ are divergent
(B) $\{a_n\}$ is convergent and $\{b_n\}$ is divergent
(C) $\{a_n\}$ is divergent and $\{b_n\}$ is convergent
(D) Both $\{a_n\}$ and $\{b_n\}$ are convergent

Question 75 (JAM 2018 • Q51 • ID: JAM_2018_Q51) NAT +2 M Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 Let $a_k = (-1)^{k-1}$, $s_n = a_1 + a_2 + \cdots + a_n$ and $\sigma_n = (s_1 + s_2 + \cdots + s_n)/n$, where $k, n \in \mathbb{N}$. Then $\lim_{n\to\infty} \sigma_n$ is _____ (correct up to one decimal place).

Question 76 (JAM 2018 • Q47 • ID: JAM_2018_Q47) NAT +1 M Neg: Nil (0)

Q.47 Let $a_n = \frac{(1+(-1)^n)}{2^n} + \frac{(1+(-1)^{n-1})}{3^n}$. Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ about $x = 0$ is _____

Question 77 (JAM 2018 • Q43 • ID: JAM_2018_Q43) NAT +1 M Neg: Nil (0)

Q.43 Let $f(x) = \sum_{n=0}^{\infty}(-1)^n x(x-1)^n$ for $0 < x < 2$. Then the value of $f\left(\frac{\pi}{4}\right)$ is _____

Question 78 (JAM 2018 • Q14 • ID: JAM_2018_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $a_n = \frac{(-1)^n}{\sqrt{1+n}}$ and let $c_n = \sum_{k=0}^{n} a_{n-k} a_k$, where $n \in \mathbb{N} \cup \{0\}$. Then which one of the following is TRUE?

(A) Both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ are convergent

(B) $\sum_{n=0}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} c_n$ is not convergent

(C) $\sum_{n=1}^{\infty} c_n$ is convergent but $\sum_{n=0}^{\infty} a_n$ is not convergent

(D) Neither $\sum_{n=0}^{\infty} a_n$ nor $\sum_{n=1}^{\infty} c_n$ is convergent

Question 79 (JAM 2018 • Q13 • ID: JAM_2018_Q13) MCQ +2 M Neg: −0.67

Q.13 Let $a_n = n + \frac{1}{n}$, $n \in \mathbb{N}$. Then the sum of the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{a_{n+1}}{n!}$ is

(A) $e^{-1} - 1$ (B) e^{-1} (C) $1 - e^{-1}$ (D) $1 + e^{-1}$

Question 80 (JAM 2018 • Q12 • ID: JAM_2018_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $a, b, c \in \mathbb{R}$. Which of the following values of a, b, c do NOT result in the convergence of the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b (\log_e n)^c} \quad ?$$

(A) $|a| < 1,\ b \in \mathbb{R}, c \in \mathbb{R}$

(B) $a = 1,\ b > 1,\ c \in \mathbb{R}$

(C) $a = 1, b \geq 0, c < 1$

(D) $a = -1,\ b \geq 0, c > 0$

Question 81 (JAM 2018 • Q11 • ID: JAM_2018_Q11) MCQ +2 M Neg: −0.67

Q. 11 – Q. 30 carry two marks each.

Q.11 Let $a_n = \begin{cases} 2 + \frac{(-1)^{\frac{n-1}{2}}}{n} & , \quad if\ n\ is\ odd \\ 1 + \frac{1}{2^n} & , \quad if\ n\ is\ even \end{cases}$, $n \in \mathbb{N}$.

Then which one of the following is TRUE?

(A) $\sup\{a_n \mid n \in \mathbb{N}\} = 3$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(B) $\liminf(a_n) = \limsup(a_n) = \frac{3}{2}$

(C) $\sup\{a_n \mid n \in \mathbb{N}\} = 2$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(D) $\liminf(a_n) = 1$ and $\limsup(a_n) = 3$

Question 82 (JAM 2018 • Q10 • ID: JAM_2018_Q10) MCQ +1 M Neg: −0.33

Q.10 Let $s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$ for $n \in \mathbb{N}$. Then which one of the following is TRUE for the sequence $\{s_n\}_{n=1}^{\infty}$

(A) $\{s_n\}_{n=1}^{\infty}$ converges in $\mathbb{Q}$

(B) $\{s_n\}_{n=1}^{\infty}$ is a Cauchy sequence but does not converge in $\mathbb{Q}$

(C) the subsequence $\{s_{k^n}\}_{n=1}^{\infty}$ is convergent in $\mathbb{R}$, only when k is even natural number

(D) $\{s_n\}_{n=1}^{\infty}$ is not a Cauchy sequence

Question 83 (JAM 2018 • Q2 • ID: JAM_2018_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $a_n = \frac{b_{n+1}}{b_n}$, where $b_1 = 1$, $b_2 = 1$ and $b_{n+2} = b_n + b_{n+1}$, $n \in \mathbb{N}$. Then $\lim_{n\to\infty} a_n$ is

(A) $\frac{1-\sqrt{5}}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{1+\sqrt{3}}{2}$ (D) $\frac{1+\sqrt{5}}{2}$

Question 84 (JAM 2017 • Q59 • ID: JAM_2017_Q59) NAT +2 M Neg: Nil (0)

Q.59 The radius of convergence of the power series

$$\sum_{0}^{\infty} n!\, x^{n^2}$$

is _____

Question 85 (JAM 2017 • Q55 • ID: JAM_2017_Q55) NAT +2 M Neg: Nil (0)

Q.55 Let $a_n = \sqrt{n}$, $n \geq 1$, and let $s_n = a_1 + a_2 + \cdots + a_n$. Then

$$\lim_{n\to\infty} \left(\frac{a_n / s_n}{-\ln(1 - a_n/s_n)} \right) = _____$$

Question 86 (JAM 2017 • Q47 • ID: JAM_2017_Q47) NAT +1 M Neg: Nil (0)

Q.47

$$\frac{1}{2\pi}\left(\frac{\pi^3}{1!\,3} - \frac{\pi^5}{3!\,5} + \frac{\pi^7}{5!\,7} - \cdots + \frac{(-1)^{n-1}\pi^{2n+1}}{(2n-1)!\,(2n+1)} + \cdots \right) = _____$$

Question 87 (JAM 2017 • Q36 • ID: JAM_2017_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Let $\{x_n\}$ be a real sequence such that $7x_{n+1} = x_n^3 + 6$ for $n \geq 1$. Then which of the following statements is/are TRUE?

(A) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 1

(B) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 2

(C) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to 1

(D) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to -3

Question 88 (JAM 2017 • Q28 • ID: JAM_2017_Q28) MCQ +2 M Neg: −0.67

Q.28 Which one of the following is TRUE?

(A) Every sequence that has a convergent subsequence is a Cauchy sequence

(B) Every sequence that has a convergent subsequence is a bounded sequence

(C) The sequence $\{\sin n\}$ has a convergent subsequence

(D) The sequence $\left\{n\cos\frac{1}{n}\right\}$ has a convergent subsequence

Question 89 (JAM 2017 • Q27 • ID: JAM_2017_Q27) MCQ +2 M Neg: −0.67

Q.27

$$\lim_{n\to\infty}\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{3}+\sqrt{6}}+\frac{1}{\sqrt{6}+\sqrt{9}}+\cdots+\frac{1}{\sqrt{3n}+\sqrt{3n+3}}\right)=$$

(A) $1+\sqrt{3}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{1+\sqrt{3}}$

Question 90 (JAM 2017 • Q26 • ID: JAM_2017_Q26) MCQ +2 M Neg: −0.67

Q.26 Let $0 < a_1 < b_1$. For $n \geq 1$, define

$$a_{n+1} = \sqrt{a_n b_n} \quad \text{and} \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

Then which one of the following is NOT TRUE?

(A) Both $\{a_n\}$ and $\{b_n\}$ converge, but the limits are not equal

(B) Both $\{a_n\}$ and $\{b_n\}$ converge and the limits are equal

(C) $\{b_n\}$ is a decreasing sequence

(D) $\{a_n\}$ is an increasing sequence

Question 91 (JAM 2017 • Q21 • ID: JAM_2017_Q21) MCQ +2 M Neg: −0.67

Q.21 The interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{1}{(-3)^{n+2}} \frac{(4x-12)^n}{n^2+1}$$

is

(A) $\frac{10}{4} \leq x < \frac{14}{4}$

(B) $\frac{9}{4} \leq x < \frac{15}{4}$

(C) $\frac{10}{4} \leq x \leq \frac{14}{4}$

(D) $\frac{9}{4} \leq x \leq \frac{15}{4}$

Question 92 (JAM 2017 • Q13 • ID: JAM_2017_Q13) MCQ +2 M Neg: −0.67

Q.13 Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then $\lim_{n\to\infty} M^n x$

(A) does not exist

(B) is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(C) is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

(D) is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Question 93 (JAM 2017 • Q9 • ID: JAM_2017_Q9) MCQ +1 M Neg: −0.33

Q.9 If $\lim_{T\to\infty}\int_0^T e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$, then

$$\lim_{T\to\infty}\int_0^T x^2 e^{-x^2}\,dx =$$

(A) $\frac{\sqrt{\pi}}{4}$ (B) $\frac{\sqrt{\pi}}{2}$ (C) $\sqrt{2\pi}$ (D) $2\sqrt{\pi}$

Question 94 (JAM 2017 • Q4 • ID: JAM_2017_Q4) MCQ +1 M Neg: −0.33

Q.4

$$\lim_{n\to\infty}\frac{\pi}{n}\sum_{k=1}^{n}\sin\left(\frac{\pi}{2}+\frac{5\pi}{2}\cdot\frac{k}{n}\right)=$$

(A) $\frac{2\pi}{5}$ (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{5\pi}{2}$

Question 95 (JAM 2016 • Q51 • ID: JAM_2016_Q51) NAT +2 M Neg: Nil (0)

Q.51 The value of $\lim_{n\to\infty}\left(8n-\frac{1}{n}\right)^{\frac{(-1)^n}{n^2}}$ is equal to _____

Question 96 (JAM 2016 • Q48 • ID: JAM_2016_Q48) NAT +1 M Neg: Nil (0)

Q.48 The radius of convergence of the power series

$$\sum_{n=1}^{\infty}\frac{(-4)^n}{n(n+1)}(x+2)^{2n}$$ is _____

Question 97 (JAM 2016 • Q42 • ID: JAM_2016_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let $\{s_k\}$ be a sequence of real numbers, where

$$s_k = k^{\alpha/k}, \qquad k \geq 1, \qquad \alpha > 0.$$

Then

$$\lim_{n\to\infty} \left(s_1\ s_2\ \ldots\ s_n\right)^{1/n}$$

is _____

Question 98 (JAM 2016 • Q41 • ID: JAM_2016_Q41) NAT +1 M Neg: Nil (0)

Q.41 Let $\{s_n\}$ be a sequence of real numbers given by

$$s_n = 2^{(-1)^n}\left(1-\frac{1}{n}\right)\sin\frac{n\pi}{2}, \qquad n \in \mathbb{N}.$$

Then the least upper bound of the sequence $\{s_n\}$ is _____

Question 99 (JAM 2016 • Q31 • ID: JAM_2016_Q31) MSQ +2 M Neg: Nil (0)

Q.31 Let $\{s_n\}$ be a sequence of positive real numbers satisfying

$$2\,s_{n+1} = s_n^2 + \frac{3}{4}, \qquad n \geq 1.$$

If α and β are the roots of the equation $x^2 - 2x + \frac{3}{4} = 0$ and $\alpha < s_1 < \beta$, then which of the following statement(s) is(are) TRUE ?

(A) $\{s_n\}$ is monotonically decreasing
(B) $\{s_n\}$ is monotonically increasing
(C) $\lim_{n\to\infty} s_n = \alpha$
(D) $\lim_{n\to\infty} s_n = \beta$

Question 100 (JAM 2016 • Q24 • ID: JAM_2016_Q24) MCQ +2 M Neg: −0.67

Q.24 The sum of the series

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2+n-2}$$

is

(A) $\frac{1}{3}\ln 2 - \frac{5}{18}$ (B) $\frac{1}{3}\ln 2 - \frac{5}{6}$ (C) $\frac{2}{3}\ln 2 - \frac{5}{18}$ (D) $\frac{2}{3}\ln 2 - \frac{5}{6}$

Question 101 (JAM 2016 • Q12 • ID: JAM_2016_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $\{a_n\}$ be a sequence of positive real numbers satisfying

$$\frac{4}{a_{n+1}} = \frac{3}{a_n} + \frac{a_n^3}{81}, \qquad n \geq 1, \ a_1 = 1.$$

Then all the terms of the sequence lie in

(A) $\left[\frac{1}{2}, \frac{3}{2}\right]$ (B) $[0,\ 1]$ (C) $[1,\ 2]$ (D) $[1,\ 3]$

Question 102 (JAM 2016 • Q11 • ID: JAM_2016_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S be the series

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)\,2^{(2k-1)}}$$

and T be the series

$$\sum_{k=2}^{\infty} \left(\frac{3k-4}{3k+2}\right)^{\frac{(k+1)}{3}}$$

of real numbers. Then which one of the following is TRUE?

(A) Both the series S and T are convergent
(B) S is convergent and T is divergent
(C) S is divergent and T is convergent
(D) Both the series S and T are divergent

Question 103 (JAM 2016 • Q1 • ID: JAM_2016_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 The sequence $\{s_n\}$ of real numbers given by

$$s_n = \frac{\sin\frac{\pi}{2}}{1\cdot 2} + \frac{\sin\frac{\pi}{2^2}}{2\cdot 3} + \cdots + \frac{\sin\frac{\pi}{2^n}}{n\cdot(n+1)}$$

is

(A) a divergent sequence
(B) an oscillatory sequence
(C) not a Cauchy sequence
(D) a Cauchy sequence

Question 104 (JAM 2015 • Q57 • ID: JAM_2015_Q57) NAT +2 M Neg: Nil (0)

Q.17

If $\int_0^x \left(e^{-t^2} + \cos t\right)\, dt$ has the power series expansion $\sum_{n=1}^{\infty} a_n x^n$, then a_5, correct upto three decimal places, is equal to _____

Question 105 (JAM 2015 • Q46 • ID: JAM_2015_Q46) NAT +1 M Neg: Nil (0)

Q.6

If the power series

$$\sum_{n=0}^{\infty} \frac{n!}{n^n} x^{2n}$$

converges for $|x| < c$ and diverges for $|x| > c$, then the value of c, correct upto three decimal places, is _____

Question 106 (JAM 2015 • Q33 • ID: JAM_2015_Q33) MSQ +2 M Neg: Nil (0)

Q.3

Which of the following conditions implies (imply) the convergence of a sequence $\{x_n\}$ of real numbers?

(A) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $|x_{n+1} - x_n| < \varepsilon$

(B) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $\frac{1}{(n+1)^2}|x_{n+1} - x_n| < \varepsilon$

(C) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $(n+1)^2|x_{n+1} - x_n| < \varepsilon$

(D) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all m, n with $m > n \geq n_0$, $|x_m - x_n| < \varepsilon$

Question 107 (JAM 2015 • Q23 • ID: JAM_2015_Q23) MCQ +2 M Neg: −0.67

Q.23

The sequence $\left\{\cos\left(\frac{1}{2}\tan^{-1}\left(-\frac{n}{2}\right)^n\right)\right\}$ is

(A) monotone and convergent
(B) monotone but not convergent
(C) convergent but not monotone
(D) neither monotone nor convergent

Question 108 (JAM 2015 • Q22 • ID: JAM_2015_Q22) MCQ +2 M Neg: −0.67

Q.22

Which one of the following statements is true for the series $\sum_{n=1}^{\infty}(-1)^n\frac{(2n)!}{n^{2n}}$?

(A) The series converges conditionally but not absolutely
(B) The series converges absolutely
(C) The sequence of partial sums of the series is bounded but not convergent
(D) The sequence of partial sums of the series is unbounded

Question 109 (JAM 2015 • Q5 • ID: JAM_2015_Q5) MCQ +1 M Neg: −0.33

Q.5 Let $\{x_n\}$ be a convergent sequence of real numbers. If $x_1 > \pi + \sqrt{2}$ and $x_{n+1} = \pi + \sqrt{x_n - \pi}$ for $n \geq 1$, then which one of the following is the limit of this sequence?

(A) $\pi + 1$ (B) $\pi + \sqrt{2}$ (C) π (D) $\pi + \sqrt{\pi}$

Question 110 (JAM 2014 • Q19 • ID: JAM_2014_Q19) MCQ +2 M Neg: −0.67

Q.19 Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series, where $a_n = \frac{(-1)^n n}{2^n}$, $b_n = \frac{(-1)^n}{\log(n+1)}$ for all $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent

(B) $\sum_{n=1}^{\infty} a_n$ is absolutely convergent but $\sum_{n=1}^{\infty} b_n$ is conditionally convergent

(C) $\sum_{n=1}^{\infty} a_n$ is conditionally convergent but $\sum_{n=1}^{\infty} b_n$ is absolutely convergent

(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are conditionally convergent

Question 111 (JAM 2014 • Q18 • ID: JAM_2014_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $\{x_n\}$ be a sequence of real numbers such that $\lim_{n\to\infty}(x_{n+1} - x_n) = c$, where c is a positive real number. Then the sequence $\left\{\frac{x_n}{n}\right\}$

(A) is NOT bounded (B) is bounded but NOT convergent

(C) converges to c (D) converges to 0

Question 112 (JAM 2014 • Q17 • ID: JAM_2014_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $x_n = 2^{2n}\left(1-\cos\left(\frac{1}{2^n}\right)\right)$ for all $n \in \mathbb{N}$. Then the sequence $\{x_n\}$

(A) does NOT converge

(B) converges to 0

(C) converges to $\frac{1}{2}$

(D) converges to $\frac{1}{4}$

Question 113 (JAM 2014 • Q6 • ID: JAM_2014_Q6) MCQ +1 M Neg: −0.33

Q.6 The radius of convergence of the power series $\sum_{n=0}^{\infty} 2^{2n} x^{n^2}$ is

(A) $\frac{1}{4}$ (B) 1 (C) 2 (D) 4

Question 114 (JAM 2013 • Q7 • ID: JAM_2013_Q7) MCQ +2 M Neg: −0.67

Let

$$x_n = \left(1-\frac{1}{3}\right)^2\left(1-\frac{1}{6}\right)^2\left(1-\frac{1}{10}\right)^2 \cdots \left(1-\frac{1}{\frac{n(n+1)}{2}}\right)^2, \quad n \geq 2.$$

Then $\lim_{n\to\infty} x_n$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{81}$ (D) 0

Question 115 (JAM 2012 • Q13 • ID: JAM_2012_Q13) MCQ +6 M Neg: −2.0

Q.13 If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 3$, then the series $\sum_{n=0}^{\infty} a_n x^n$

(A) converges absolutely for $x = -2$
(B) converges but not absolutely for $x = -1$
(C) converges but not absolutely for $x = 1$
(D) diverges for $x = -2$

Question 116 (JAM 2012 • Q1 • ID: JAM_2012_Q1) MCQ +6 M Neg: −2.0

Q.1 Let $\{x_n\}$ be the sequence $+\sqrt{1}, -\sqrt{1}, +\sqrt{2}, -\sqrt{2}, +\sqrt{3}, -\sqrt{3}, +\sqrt{4}, -\sqrt{4}, \ldots$.
If

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n} \text{ for all } n \in \mathbb{N},$$

then the sequence $\{y_n\}$ is

(A) monotonic
(B) NOT bounded
(C) bounded but NOT convergent
(D) convergent

Question 117 (JAM 2011 • Q14 • ID: JAM_2011_Q14) MCQ +6 M Neg: −2.0

Q.14 The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{n}{(2n+1)^2} (x-2)^{3n}$ converges is

(A) $[-1, 1)$ (B) $[-1, 1]$ (C) $[1, 3)$ (D) $[1, 3]$

Question 118 (JAM 2011 • Q3 • ID: JAM_2011_Q3) MCQ +6 M Neg: −2.0

Q.3 The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2 + kn}}$ is

(A) $2(\sqrt{2} - 1)$ (B) $2\sqrt{2} - 1$ (C) $2 - \sqrt{2}$ (D) $\frac{1}{2}(\sqrt{2} - 1)$

Question 119 (JAM 2011 • Q1 • ID: JAM_2011_Q1) MCQ +6 M Neg: −2.0

Q.1 Let $a_n = \sum_{k=1}^{n} \frac{n}{n^2+k}$, for $n \in \mathbb{N}$. Then the sequence $\{a_n\}$ is

(A) Convergent
(B) Bounded but not convergent
(C) Diverges to ∞
(D) Neither bounded nor diverges to ∞

Question 120 (JAM 2010 • Q13 • ID: JAM_2010_Q13) MCQ +6 M Neg: −2.0

Q.13 The radius of convergence of the power series $\sum_{n=0}^{\infty} a_n z^{n^2}$, where $a_0 = 1$, $a_n = 3^{-n} a_{n-1}$ for $n \in \mathbf{N}$, is

(A) 0 (B) $\sqrt{3}$ (C) 3 (D) ∞

Question 121 (JAM 2010 • Q1 • ID: JAM_2010_Q1) MCQ +6 M Neg: −2.0

Q.1 Which of the following conditions does NOT ensure the convergence of a real sequence $\{a_n\}$?

(A) $|a_n - a_{n+1}| \to 0$ as $n \to \infty$
(B) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent
(C) $\sum_{n=1}^{\infty} n a_n$ is convergent
(D) The sequences $\{a_{2n}\}$, $\{a_{2n+1}\}$ and $\{a_{3n}\}$ are convergent

Question 122 (JAM 2009 • Q9 • ID: JAM_2009_Q9) MCQ +6 M Neg: −2.0

Q.9 The set of all positive values of a for which the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \tan^{-1}\left(\frac{1}{n}\right)\right)^a$ converges, is

(A) $\left(0, \frac{1}{3}\right]$ (B) $\left(0, \frac{1}{3}\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $\left(\frac{1}{3}, \infty\right)$

Question 123 (JAM 2009 • Q7 • ID: JAM_2009_Q7) MCQ +6 M Neg: −2.0

Q.7 The set of all limit points of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{9}{16}, \ldots$ is

(A) $[0, 1]$
(B) $(0, 1]$
(C) the set of all rational numbers in $[0, 1]$
(D) the set of all rational numbers in $[0, 1]$ of the form $\frac{m}{2^n}$ where m and n are integers

Question 124 (JAM 2009 • Q6 • ID: JAM_2009_Q6) MCQ +6 M Neg: −2.0

Q.6 Let $a_n = \begin{cases} \frac{1}{3^n} & \text{if } n \text{ is a prime,} \\ \frac{1}{4^n} & \text{if } n \text{ is not a prime.} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

(A) 4 (B) 3 (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Question 125 (JAM 2008 • Q12 • ID: JAM_2008_Q12) MCQ +6 M Neg: −2.0

Q.12 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1 = 1$ and for $n \geq 1$, $a_{n+1} = a_n + (-1)^n 2^{-n}$, $b_n = \frac{2a_{n+1} - a_n}{a_n}$. Then

(A) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence
(B) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence
(C) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence
(D) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence

Question 126 (JAM 2008 • Q3 • ID: JAM_2008_Q3) MCQ +6 M Neg: −2.0

Q.3 Let $s_n = \int_0^1 \frac{n x^{n-1}}{(1+x)} dx$ for $n \geq 1$. Then as $n \to \infty$, the sequence $\{s_n\}$ tends to

(A) 0 (B) 1/2 (C) 1 (D) $+\infty$

Question 127 (JAM 2007 • Q10 • ID: JAM_2007_Q10) MCQ +6 M Neg: −2.0

10. Suppose (c_n) is a sequence of real numbers such that $\lim_{n\to\infty} |c_n|^{1/n}$ exists and is non-zero. If the radius of convergence of the power series $\sum_{n=0}^{\infty} c_n x^n$ is equal to r, then the radius of convergence of the power series $\sum_{n=1}^{\infty} n^2 c_n x^n$ is

(A) less than r
(B) greater than r
(C) equal to r
(D) equal to 0

Question 128 (JAM 2007 • Q7 • ID: JAM_2007_Q7) MCQ +6 M Neg: −2.0

7. Let (a_n) be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is divergent. Let $s_n = \sum_{k=1}^{n} a_k$ for $n = 1,2,\ldots$ and $t_n = \sum_{k=2}^{n} \frac{a_k}{s_{k-1}s_k}$ for $n = 2,3,\ldots$. Then $\lim_{n\to\infty} t_n$ is equal to

(A) $1/a_1$
(B) 0
(C) $1/(a_1+a_2)$
(D) $a_1 + a_2$

Question 129 (JAM 2006 • Q1 • ID: JAM_2006_Q1) MCQ +6 M Neg: −2.0

1. $\lim_{n\to\infty} \frac{2^{n+1}+3^{n+1}}{2^n+3^n}$ equals

(A) 3
(B) 2
(C) 1
(D) 0

race

Question 130 (JAM 2005 • Q14 • ID: JAM_2005_Q14) MCQ +6 M Neg: −2.0

14. Let $f:(-1,1)\longrightarrow \boldsymbol{R}$ be such that $f^{(n)}(x)$ exists and $\left|f^{(n)}(x)\right|\leq 1$ for every $n\geq 1$ and for every $x\in(-1,1)$. Then f has a convergent power series expansion in a neighbourhood of

(A) every $x\in(-1,1)$

(B) every $x\in\left(\frac{-1}{2},0\right)$ only

(C) no $x\in(-1,1)$

(D) every $x\in\left(0,\frac{1}{2}\right)$ only

Question 131 (JAM 2005 • Q1 • ID: JAM_2005_Q1) MCQ +6 M Neg: −2.0

1. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n=a_{2n}$ and $c_n=a_{2n+1}$. Then $\{a_n\}$ is convergent

(A) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent

(B) implies $\{c_n\}$ is convergent but $\{b_n\}$ neet not be convergent

(C) implies both $\{b_n\}$ and $\{c_n\}$ are convergent

(D) if both $\{b_n\}$ and $\{c_n\}$ are convergent

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+1	1.0 to 1.0
2	NAT	+1	1.0 to 1.0
3	NAT	+1	4.0 to 4.0
4	MSQ	+2	C;D
5	MCQ	+2	A
6	MCQ	+2	C
7	MCQ	+2	C
8	MCQ	+1	D
9	MCQ	+1	D
10	NAT	+2	5.20 to 5.24
11	NAT	+1	1.98 to 2.02
12	MSQ	+2	A;C;D
13	MCQ	+2	B
14	MCQ	+2	A
15	MCQ	+2	B
16	MCQ	+2	A
17	MCQ	+2	D
18	MCQ	+2	D
19	MCQ	+1	C
20	NAT	+2	0.98 to 1.02
21	NAT	+2	0.30 to 0.40
22	NAT	+1	1.95 to 2.05
23	MSQ	+2	B;C
24	MSQ	+2	A;B
25	MSQ	+2	A;B
26	MCQ	+2	B
27	MCQ	+2	A
28	MCQ	+2	A
29	MCQ	+2	D
30	NAT	+1	0.99 to 1.01
31	NAT	+1	0.49 to 0.51
32	MSQ	+2	A, D
33	MCQ	+2	B
34	MCQ	+2	B
35	MCQ	+2	C
36	MCQ	+2	B
37	MCQ	+2	B
38	MCQ	+1	C
39	MCQ	+1	C
45	MCQ	+2	C
46	MCQ	+1	A
47	MCQ	+1	C
48	MCQ	+1	D
49	MCQ	+1	B
50	NAT	+2	0
51	NAT	+1	1
52	NAT	+1	7
53	MCQ	+2	B
54	MCQ	+2	B
55	NAT	+2	33 to 33
56	NAT	+2	0.25 to 0.25
57	NAT	+1	1 to 1
58	MSQ	+2	A;D
59	MSQ	+2	B;C
60	MSQ	+2	A;D
61	MCQ	+2	D
62	MCQ	+2	A
63	MCQ	+2	D
64	MCQ	+1	D
65	MCQ	+1	C
66	NAT	+2	2.69 to 2.74
67	MSQ	+2	A, B
68	MSQ	+2	A
69	MCQ	+2	A
70	MCQ	+2	C
71	MCQ	+2	B
72	MCQ	+2	A
73	MCQ	+1	A
74	MCQ	+1	D
75	NAT	+2	0.4 to 0.6
76	NAT	+1	2 to 2
77	NAT	+1	1 to 1
78	MCQ	+2	B
79	MCQ	+2	D
80	MCQ	+2	C
81	MCQ	+2	A
82	MCQ	+1	B
83	MCQ	+1	D
89	MCQ	+2	C
90	MCQ	+2	A
91	MCQ	+2	D
92	MCQ	+2	C
93	MCQ	+1	A
94	MCQ	+1	C
95	NAT	+2	1.0 to 1.0
96	NAT	+1	0.5 to 0.5
97	NAT	+1	1.0 to 1.0
98	NAT	+1	0.5 to 0.5
99	MSQ	+2	A;C
100	MCQ	+2	C
101	MCQ	+2	D
102	MCQ	+2	B
103	MCQ	+1	D
104	NAT	+2	0.10 to 0.11
105	NAT	+1	1.64 to 1.65
106	MSQ	+2	C;D
107	MCQ	+2	A
108	MCQ	+2	B
109	MCQ	+1	A
110	MCQ	+2	B
111	MCQ	+2	C
112	MCQ	+2	C
113	MCQ	+1	B
114	MCQ	+2	B
115	MCQ	+6	A
116	MCQ	+6	D
117	MCQ	+6	C
118	MCQ	+6	A
119	MCQ	+6	A
120	MCQ	+6	B
121	MCQ	+6	A
122	MCQ	+6	C
123	MCQ	+6	A
124	MCQ	+6	B
125	MCQ	+6	B
126	MCQ	+6	B
127	MCQ	+6	C

Q. No.	Type	Marks	Official Key
40	NAT	+2	**3.00 to 3.00**
41	NAT	+1	**0.24 to 0.26**
42	MSQ	+2	**A,C**
43	MCQ	+2	**B**
44	MCQ	+2	**D**

Q. No.	Type	Marks	Official Key
84	NAT	+2	**0.9 to 1.1**
85	NAT	+2	**0.9 to 1.1**
86	NAT	+1	**0.49 to 0.51**
87	MSQ	+2	**A, C**
88	MCQ	+2	**C**

Q. No.	Type	Marks	Official Key
128	MCQ	+6	**A**
129	MCQ	+6	**A**
130	MCQ	+6	**A**
131	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.